

Fridge Recipe Tracker — Engineering Report

Team Project Part 1 · Programming Fundamentals · AI-assisted, food-waste-aware recipe planner (procedural Python CLI)

Lab group: [LabGroup] Team: [TeamNumber]

Submission: 14 October 2026

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Repository: github.com/cclouden56/PROJECTKITCHEN

1. Problem and Objectives

Households routinely discard food that was simply forgotten until it expired, then spend more on groceries or takeaway, and face daily "what can I cook?" decision fatigue. **Fridge Recipe Tracker** is a command-line kitchen assistant that records fridge and pantry items with expiry dates and asks Google Gemini for a recipe that uses the most urgent items first. The AI output is never trusted blindly: a deterministic rule engine accepts, flags or rejects every suggestion before it reaches the user.

- **Track** inventory (name, quantity, unit, location, expiry) in flat JSON files that survive restarts and container rebuilds.
- **Generate** recipes with AI under the user's meal type, time limit, allergies and must-include ingredients, using a strict JSON response schema.
- **Decide** with business rules: allergen gate (with a synonym map, e.g. *dairy* → *milk, cheese, butter*), ≥ 50 % ingredient-match threshold, cook-time limit, must-include check, and a minimum of two inventory items before calling the API.
- **Measure impact**: an expiry-priority score (expired/today +40, 1–3 days +30, 4–7 days +15, otherwise +5, capped at 100) and an estimate of grams of food diverted from waste.

2. Architecture

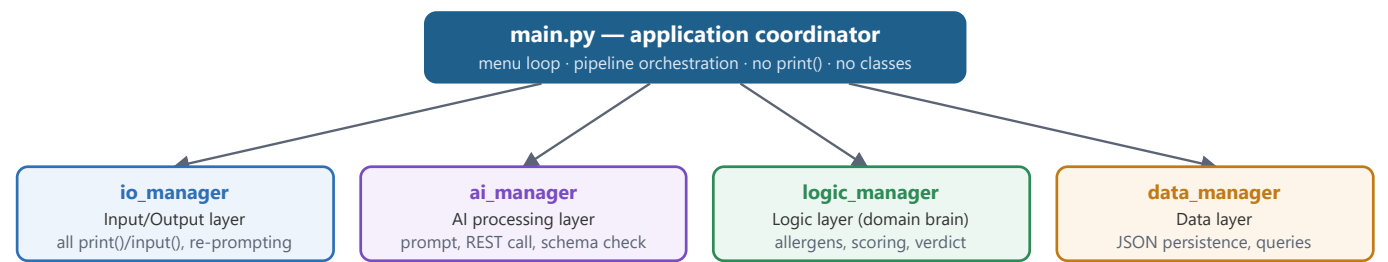


Figure 1. Hub-and-spoke modular architecture. No manager imports another manager; only `main.py` composes them, so each layer can be tested and replaced in isolation.

Module	Single responsibility	Key procedural functions (all typed)
io_manager	Collect and validate every keystroke; render every line of output. The only module allowed to call <code>print()</code> .	<code>prompt_menu_choice</code> , <code>prompt_date</code> , <code>prompt_positive_number</code> , <code>prompt_recipe_generation_criteria</code> , <code>display_inventory</code> , <code>display_recipe_card</code>
ai_manager	Build the prompt, call the Gemini REST API with a model cascade, and validate the JSON schema. Contains no domain rules.	<code>build_recipe_prompt</code> , <code>call_gemini_api</code> , <code>classify_api_error</code> , <code>extract_recipe_json</code> , <code>validate_recipe_schema</code> , <code>request_recipe_from_ai</code>
logic_manager	Apply the business rules to validated AI output and produce a verdict: ACCEPTED, FLAGGED or REJECTED.	<code>evaluate_allergen_safety</code> , <code>calculate_ingredient_match_ratio</code> , <code>calculate_expiry_priority_score</code> , <code>check_mandatory_ingredients</code> , <code>evaluate_recipe</code>
data_manager	Load, save and query flat JSON files; survive missing, corrupt or unwritable files.	<code>load_json_data</code> , <code>save_json_data</code> , <code>quarantine_corrupt_file</code> , <code>add_inventory_item</code> , <code>query_items_by_expiry</code> , <code>search_recipes_by_ingredient</code>

Design conventions. The program is 100 % procedural: there are zero `class` definitions, and an automated test scans every `.py` file to enforce this. Layers exchange only plain `dict/list` values. Failures are returned as values (`(success, result, error_message)` tuples, defaults or `False`), so an exception never crosses a layer boundary and the menu loop cannot crash. Diagnostics go to `data/app.log` through logging, keeping the terminal UI clean.

3. Data Flow: User Input → AI Payload → File Storage

Figure 2 follows a single "AI Recipe Generator" request (menu option 4) through the system. Each numbered box is a distinct **state** of the data; each arrow is the function that transforms it. Swimlanes show which layer owns each state.

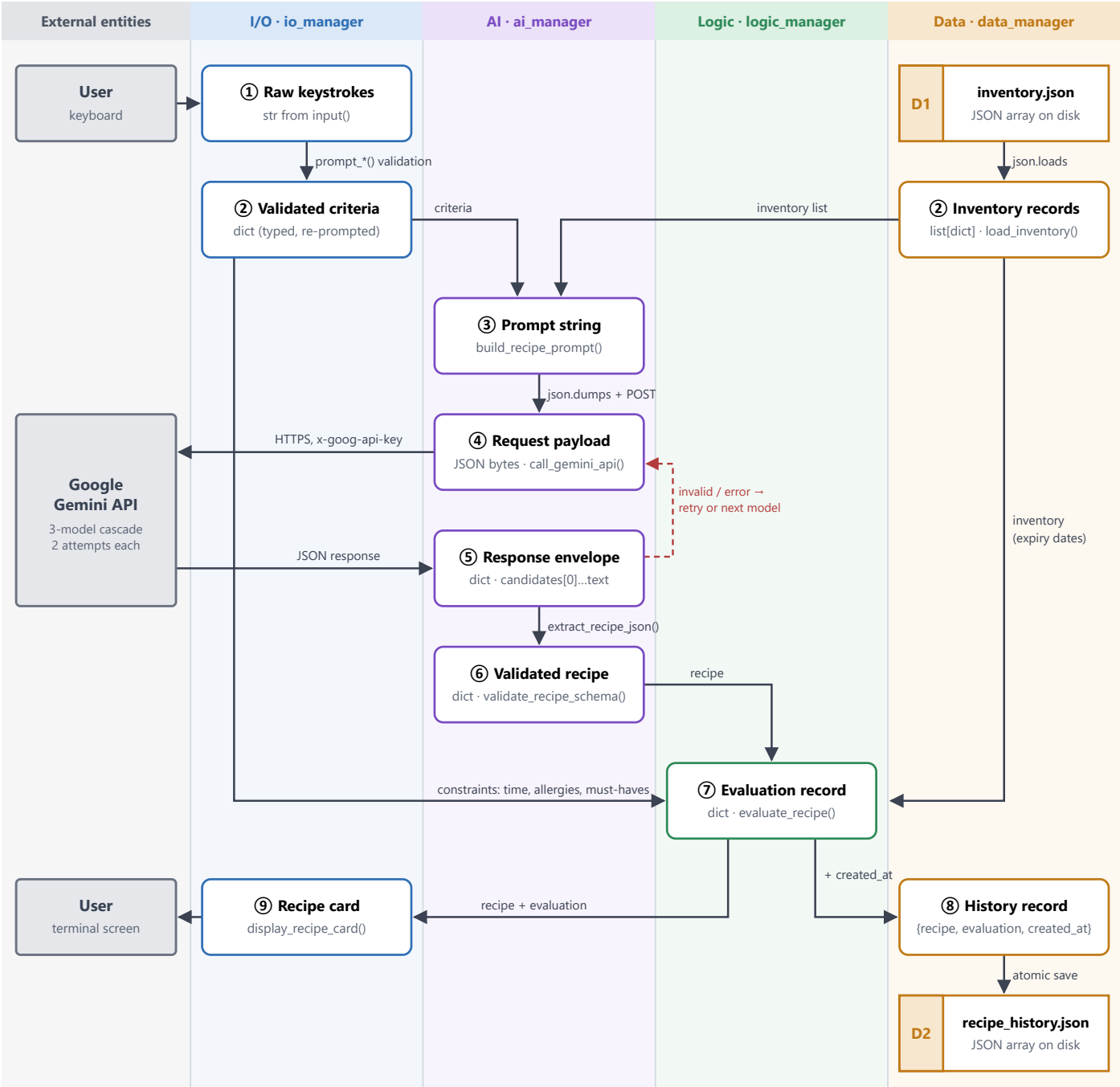


Figure 2. Data flow diagram for recipe generation. Solid arrows carry data between states; the dashed red loop is the recovery path for a failed or invalid response (retry the same model on transient errors, otherwise move to the next model). If every attempt fails, an error goes to the user, **nothing is written to D2**, and control returns to the menu.

Inventory path (menu option 2): user input → file storage

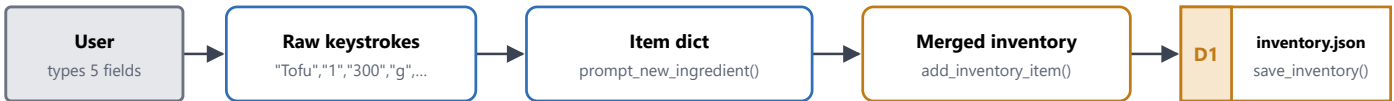


Figure 3. The quantity is coerced to a positive finite float and the date to the canonical YYYY-MM-DD string; an existing item with the same name (case-insensitive) is updated rather than duplicated, and a new item gets the next `ing-NNN` id.

State of the data at each stage

Example values come from the automated end-to-end test (T12), which drives the real CLI with scripted keystrokes and a hardcoded Gemini response.

State	Type	Transformation	Example value
① Raw keystrokes	str	input() inside io_manager only	"4", "1", "20", "nuts", "", ""
② Validated criteria	dict	prompt_recipe_generation_criteria(): typed, trimmed, comma-split	{"meal_type": "Breakfast", "max_cook_time_mins": 20, "allergies": ["nuts"], "mandatory_ingredients": [], "additional_notes": ""}
② Inventory records	list[dict]	load_inventory(): JSON → objects; non-dict rows dropped	[{"id": "ing-001", "name": "Eggs", "quantity": 6, "unit": "pcs", "location": "Fridge", "expiry_date": "2026-10-09"}, ...]
③ Prompt string	str	build_recipe_prompt(): inventory + constraints + required JSON schema	"...- Eggs (6 pcs, Location: Fridge, Expiry: 2026-10-09) ... Restrictions to avoid: nuts ... You MUST respond strictly with a valid JSON object..."
④ Request payload	bytes	json.dumps().encode(); POST, key in x-goog-api-key header	{"contents": [{"parts": [{"text": "<prompt>"}]}], "generationConfig": {"responseMimeType": "application/json", "temperature": 0.4}}
⑤ Response envelope	dict	json.loads() of the HTTP body	{"candidates": [{"content": {"parts": [{"text": "\\{\\\"recipe_name\\\": \\\"Spinach & Cheddar Omelette\\\", ...}\\\"}"}], "finishReason": "STOP"}]}
⑥ Validated recipe	dict	extract_recipe_json() → validate_recipe_schema(): 8 fields type-checked, cast, trimmed	{"recipe_name": "Spinach & Cheddar Omelette", "estimated_cook_time_mins": 15, "ingredients_used": ["3 Eggs", "50g Cheddar Cheese, shredded", "1 cup fresh Spinach"], "missing_ingredients": [], "required_tools": [...], "instructions": [...], ...}
⑦ Evaluation record	dict	evaluate_recipe(⑥, ②, constraints)	{"outcome": "ACCEPTED", "status_code": "OPTIMAL_MATCH", "match_ratio": 1.0, "expiry_priority_score": 100.0, "rescued_items": [Eggs, Cheddar Cheese, Spinach], "waste_diverted_grams": 950, ...}
⑧ History record → D2	dict → JSON	save_recipe_to_history(): append, write .tmp, atomic os.replace()	{"recipe": {⑥}, "evaluation": {⑦}, "created_at": "2026-10-08 14:52:10"} appended to data/recipe_history.json
⑨ Recipe card	text	display_recipe_card(⑥, ⑦)	Outcome Status : [ACCEPTED] - OPTIMAL_MATCH · Expiry Priority : 100.0 / 100 · Food Waste Saved: ~950 grams

How the logic layer turns ⑥ into a verdict (⑦)

#	Condition (checked in order; first match wins)	Outcome	status_code
1	Any ingredient matches a user allergen or a synonym of one	REJECTED	ALLERGEN_CONTAMINATION
2	Ingredient match ratio < 50 % (pantry staples such as salt, black pepper and cooking oil are not counted)	REJECTED	INSUFFICIENT_INGREDIENTS
3	A must-include ingredient is absent from the recipe	FLAGGED	MISSING_MANDATORY_INGREDIENT
4	Cook time exceeds the user's limit	FLAGGED	EXCEEDS_TIME_LIMIT
5	Some ingredients missing, but ratio ≥ 50 %	FLAGGED	MISSING_SOME_INGREDIENTS
6	Otherwise	ACCEPTED	OPTIMAL_MATCH

4. Exception Handling Matrix

All four required failure classes are covered. The **Test** column names the automated test (Section 5) that triggers each scenario offline and asserts the stated behaviour. No failure is allowed to crash the menu loop.

#	Scenario	Detected by	Handling & recovery → what the user sees	Test
A. API connection failure				
1	No API key configured	get_api_key() returns "" in request_recipe_from_ai	Fails fast, before any network call. → [ERROR] "Gemini API key is not configured...", then back to the menu.	T9 T12
2	Network down, DNS failure, refused connection, or 30 s timeout	URLError / TimeoutError caught in call_gemini_api	Returned as (False, None, msg) and logged; classified <i>retry</i> : 1.5 s back-off, 2 attempts × 3 models; nothing saved. → "AI API service temporarily unavailable..." if all fail.	T9
3	Rate limited or server busy (HTTP 429 / 5xx)	HTTPError → classify_api_error() = retry	Back off, retry the same model, then move down the cascade. → Invisible if a retry succeeds.	T9
4	Invalid or revoked key (401, 403, API_KEY_INVALID)	classify_api_error() = abort	Stops after one call, because no other model can succeed with a bad key. → "AI request rejected (check your GEMINI_API_KEY)".	T9 T12
5	Model retired or request rejected (404 / 400)	classify_api_error() = next_model	Skips the second attempt and moves to the next model in the cascade. → Invisible if a fallback model succeeds.	T9
B. Malformed API response				
6	HTTP body is not JSON, or JSON is not an object	JSONDecodeError / isinstance check in call_gemini_api	Counted as a failed attempt → retry, then cascade.	T9
7	No candidates (safety block), no parts, empty text, or a candidate that is not an object	Guard clauses + AttributeError / TypeError in extract_recipe_json	Returns (False, {}, reason) and never raises; the request loop tries again.	T8
8	Prose or truncated JSON instead of a JSON object	JSONDecodeError in extract_recipe_json	Rejected; the next valid response is used (a malformed-then-valid sequence is tested).	T8 T9
9	Valid JSON wrapped in ```json fences	strip_markdown_fences()	Fences are stripped and the JSON accepted (tolerant reader). → Normal recipe card.	T8
10	Missing field, wrong type ("about 10 minutes", true), negative or NaN cook time, empty name, ingredients or steps	validate_recipe_schema() (8 required fields, types, ranges)	Rejected with a field-specific reason; invalid data never reaches the logic layer. → After the final attempt: [ERROR] with the last reason; nothing saved.	T4 T8
C. Corrupt or missing data file				
11	File or data/ folder missing (first run, fresh container), or empty file	os.path.exists and an empty-content check in load_json_data	Returns [] and creates the folder; the file is created on first save. → "Your inventory is currently empty..."	T3 T10
12	Corrupt JSON (truncated or hand-edited)	JSONDecodeError in load_json_data	Logged; the file is moved to *.corrupt-<timestamp>.bak, so the next save cannot overwrite the original bytes; the app continues with [].	T3 T10
13	Wrong shape (object instead of array, non-object rows)	isinstance filters in load_inventory / load_recipe_history	Invalid rows are dropped and valid rows kept.	T10
14	Unreadable file (permissions, invalid UTF-8)	OSError / UnicodeDecodeError	Logged; the default [] is returned and the app keeps running.	—
15	Write fails (read-only disk, bad path, unserialisable value)	OSError / TypeError / ValueError in save_json_data	Returns False and removes the temp file; the caller reports it. → [ERROR] "Failed to save inventory to storage."	T10
16	Crash or power loss during a save	Prevented by design	Data is written to file.tmp and then swapped in with an atomic os.replace(), so the stored file is never half-written.	T10
D. Invalid user input				
17	Blank text; non-numeric or out-of-range menu, meal or storage choice	prompt_menu_choice, prompt_string, prompt_meal_type, prompt_storage_location	Re-prompts in a loop; the allergy list is trimmed and blank entries dropped. → "Invalid option. Please choose between 1 and 7."	T11
18	Bad or impossible date (12-10-2026, 2026-02-30)	strptime ValueError in prompt_date	Re-prompts with an example; the date is stored canonically as YYYY-MM-DD.	T11 T12
19	Zero, negative, non-numeric, inf or nan quantity or cook time	prompt_positive_number (ValueError, math.isfinite, > 0)	Re-prompts. → "Number must be strictly greater than 0." / "...a finite value."	T11 T12
20	Removing an unknown item; fewer than 2 items before an AI request	remove_inventory_item flag; rule in handle_generate_recipe	Nothing is saved and no API call is made. → [WARNING] "...was not found" / "...add more items".	T12
21	Ctrl+C or closed input stream (EOF, e.g. docker run without -it)	KeyboardInterrupt / EOFError in main()	Clean exit with code 0; every action had already been saved. → "Input stream closed. Exiting ... safely."	T12

5. Automated Testing

tests/test_pipeline.py is a procedural test runner: it defines no classes and never contacts the live API. It uses **hardcoded sample Gemini responses** (tests/sample_ai_responses.py), each a full generateContent envelope as the real API returns it, and substitutes call_gemini_api / input() with scripted stand-ins. It exits non-zero on any failure, so CI and Docker can gate on it.

ID	Test	What it proves
T1	Zero-classes constraint	A regex scan of every .py file finds no class definition
T2	print() localisation	print() appears only in src/io_manager.py (and the test reporter)
T3	Data persistence & resilience	Save/load round trip, dedupe and remove, expiry query, history filters, corrupt-file fallback
T4	AI schema validation	A valid payload is accepted; a missing field, wrong type or empty list is rejected
T5	Logic multi-condition rules	Allergen gate, expiry score, and the ACCEPTED / FLAGGED / REJECTED branches
T6	Logic rules on sample AI responses	Six hardcoded responses → extract_recipe_json → evaluate_recipe reach every verdict, with exact score (100), ratio (0.6) and waste (950 g)
T7	Logic helper functions	Fixed-date expiry maths, fuzzy matching, staples, unit conversion, allergen synonyms
T8	Malformed AI responses	Nine malformed envelopes are rejected without raising; a fenced-JSON response is accepted
T9	Offline API failure handling	Network down, timeout, non-JSON body, invalid key, 404 cascade, 429 recovery, missing key, real refused socket
T10	Corrupt & unwritable files	Quarantine to .bak, empty and wrong-shape files, atomic save, write failure returns False
T11	I/O re-prompting	Every prompt rejects blank, text, out-of-range, bad-date and inf/nan input until a valid value is given
T12	End-to-end pipeline (offline)	Drives the real CLI: add item → JSON file; remove unknown item → no change; recipe → history record; API 401 → nothing saved; EOF → clean exit

Result: 12/12 pass locally (Python 3.14, Windows 11) and inside the Docker image (Python 3.12-slim, Linux), with **no API key and no network**: docker compose run --rm test.

6. DevOps: Version Control, CI and Docker

Git workflow

- Work happens on feature branches (e.g. feature/part1-deliverables) and is merged into main by pull request once CI passes.
- One logical change per commit, with a descriptive message, e.g. *"quarantine corrupt data files and make JSON saves atomic"*.
- .env is git-ignored (only .env.example is committed), and the API key is sent in a header, never in a URL.

Docker

- Dockerfile sits in the repository root and builds on python:3.12-slim with zero third-party packages (standard library only), giving a 43 MB compressed image.
- /app/data is a volume, so data persists between runs.
docker compose run --rm app starts the app; the test service needs no .env.

CI pipeline (.github/workflows/ci.yml)

GitHub Actions runs on every push and on every pull request into main:

1. **Lint & test** on Python 3.10 and 3.12: ruff (syntax errors, undefined names), byte-compile, then the full offline suite with an empty API key.
2. **Docker**, after step 1 passes: build the image, run the suite *inside* the container, and smoke-test that the CLI starts and exits cleanly.

A red job blocks the merge, so main always builds, passes its tests and runs in a container.

7. Known Limitations

- Ingredient matching is lexical (substring and plural stemming), so "coconut milk" matches "milk". Semantic or AI-assisted matching would be more precise.
- The waste figure is a heuristic: about 100 g per countable unit, with the whole stock of each rescued item counted. Quantities are not decremented after cooking.
- Pantry staples do not lower the match ratio, but they are still listed as missing and can therefore produce an informational FLAGGED verdict.